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Original Research Article

Performance and Interpretability Trade-Offs in Reinforcement Learning for Sepsis Treatment: Comparing Offline and Online Approaches

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Abstract

Background/Objectives: Sepsis is a leading cause of mortality in critical care and is associated with substantial long-term neurological morbidity among survivors. Reinforcement learning offers a route to data-driven treatment policies, but clinical adoption is limited by concerns that interpretability must be sacrificed for performance and that interactive (online) training is inherently superior to learning from static (offline) data. We benchmark these assumptions by comparing outcome performance and quantitative interpretability across offline and online reinforcement learning approaches for sepsis treatment. **Methods**: Using the gym-sepsis simulator derived from intensive care electronic health records, we trained offline policies with behavior cloning and conservative Q-learning on a fixed dataset of 2,105 simulated episodes (approximately 20,000 transitions) generated by a guideline-inspired heuristic policy. We also trained three online methods (double deep Q-network with attention, double deep Q-network with residual connections, and soft actor-critic) with simulator interaction, and included a vanilla deep Q-network as a baseline trained via interaction. Policies were evaluated on 500 newly sampled episodes per method, reporting overall survival and survival stratified by initial Sequential Organ Failure Assessment severity. Interpretability was quantified via linearly estimated gradients feature-saliency analysis. **Results**: Survival rates were tightly clustered (94.0%–95.4%) across learned policies, and the random baseline also achieved high survival, indicating limited outcome sensitivity within the simulator. For high-severity episodes (Sequential Organ Failure Assessment score ≥ 11), the attention-based online method reached 90.5% survival, compared with 88.5%–88.6% for conservative Q-learning and behavior cloning, while the vanilla deep Q-network underperformed (84.3%). Interpretability differed substantially: conservative Q-learning produced the strongest saliency signals (maximum saliency 40.06, emphasizing clinically relevant hemodynamic features such as systolic blood pressure), online methods showed intermediate magnitudes (maximum 1.17–3.57), behavior cloning was lower (0.78), and the vanilla deep Q-network was negligible (0.069). **Conclusions**: In this sepsis treatment simulator, online methods provided at most marginal survival improvements, while conservative offline learning achieved comparable outcomes with markedly stronger and more clinically coherent feature-importance signals and without requiring exploratory interaction during training. These findings support prioritizing conservative offline approaches when safety and explainability are central, while emphasizing that generalization beyond simulation requires prospective validation.

Keywords: Reinforcement learning, Sepsis treatment, Interpretability, Conservative Q-learning, LEG analysis, Offline RL, MIMIC-III, Sepsis-associated encephalopathy, Neurological outcomes, Blood-brain barrier, AI in neurology

1. Introduction

Sepsis remains one of the most pressing challenges in critical care, responsible for nearly 20% of global mortality and more than $62 billion in annual United States healthcare expenditures.[1,2] Despite successive iterations of the Sepsis-3 definition[3] and aggressive early-intervention campaigns, outcomes have plateaued. Mortality still ranges from 10–20% for sepsis without shock to 40–50% for septic shock. Clinicians must synthesize heterogeneous physiological signals and act within hours, yet existing protocols offer only population-level heuristics for fluid resuscitation,

vasopressor titration, and escalation to organ support.[4] Large randomized trials that re-evaluated early goal-directed therapy[5,6] underscore the difficulty of prescribing universally optimal intervention thresholds.

Sepsis has particularly profound implications for patients with neurological disorders. The systemic inflammatory response characteristic of sepsis can accelerate neurodegenerative processes through multiple mechanisms, including blood–brain barrier (BBB) disruption, microglial activation, and cytokine-mediated neuronal damage.[7,8] In patients with pre-existing conditions such as Alzheimer's disease, sepsis episodes are associated with accelerated cognitive decline and increased mortality.[9] This bidirectional relationship, where neurological disorders increase sepsis vulnerability while sepsis exacerbates neurodegeneration, underscores the critical need for optimized, interpretable treatment strategies that clinicians can confidently apply in these complex cases.

Reinforcement learning (RL) has emerged as a candidate framework for tailoring sepsis therapy to patient trajectories. By optimizing long-horizon rewards, RL-based policies can, in principle, balance competing short-term hemodynamic targets against downstream survival. Early work trained a Deep Q-Network (DQN) and fitted Q-iteration policies on Medical Information Mart for Intensive Care III (MIMIC-III) data, showing promising retrospective survival estimates.[10,11] However, these studies emphasized expected returns and policy deviations from clinician behavior while offering only qualitative or aggregate descriptions of why certain actions were recommended. As sepsis RL research shifts toward the offline setting where algorithms must learn exclusively from historical data, questions about policy interpretability become central. Offline methods ranging from Behavior Cloning (BC) to Conservative Q-Learning (CQL) and offline-adapted DQN handle uncertainty and distribution shift differently, which plausibly shapes the transparency of their learned decision rules.

A rigorous understanding of how offline RL algorithms trade off performance and interpretability is still missing. Existing evaluations lack quantitative feature-attribution analyses, a gap with direct regulatory consequences. The United States Food and Drug Administration requires explainable artificial intelligence (AI) systems for medical decision support[12], yet no sepsis RL study has demonstrated whether high-performing policies expose clinically plausible decision rationales that clinicians can validate and trust.[13] Moreover, interpretability techniques for sequential decision-making, such as Linearly Estimated Gradients (LEG) saliency maps[14], have rarely been applied to healthcare RL. It remains unclear which algorithmic choices yield explanations aligned with accepted sepsis physiology.

Responding to this gap, we ask: Can offline RL algorithms for sepsis simultaneously deliver state-of-the-art survival performance and clinically interpretable decision rationales? We hypothesize that performance-interpretability trade-offs depend on algorithmic design and that conservatism in the objective, as in CQL, can enhance both safety and transparency. To test this hypothesis, we train BC, CQL, and DQN policies on a dataset of simulated patient trajectories generated by rolling out a heuristic policy in the gym-sepsis simulator, an environment whose dynamics and outcome models were trained on MIMIC-III data.[10] We jointly evaluate survival outcomes stratified by Sequential Organ Failure Assessment (SOFA) scores and LEG-based feature saliency, ensuring that both performance and interpretability are quantified rigorously.

This study makes the following contributions:

- To our knowledge, among the first systematic LEG-based benchmarks comparing offline and

online RL algorithms on both outcome metrics and gradient-based interpretability for sepsis management within the gym-sepsis simulation environment.

- Empirical evidence that, in this specific simulator, performance and interpretability need not trade off. CQL achieves statistically indistinguishable survival compared to other methods while exhibiting substantially stronger feature importance signals and guideline-consistent saliency patterns.
- Methodological guidance on applying LEG analysis to safety-critical RL deployments, with appropriate caveats about generalization beyond simulation to real clinical settings.

The remainder of this paper is organized as follows. **Section 2** surveys clinical RL and interpretability research. **Section 3** formalizes the sepsis decision process and LEG framework. **Section 4** details experimental design. **Section 5** reports performance and interpretability findings. **Section 6** interprets the implications for clinical adoption. **Section 7** outlines future research directions.

2. Materials and Methods

We describe the experimental framework, including the gym-sepsis simulation environment, the offline and online reinforcement learning (RL) algorithms, the LEG-based interpretability analysis, and the evaluation protocol.

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2.1 Materials

2.1.1 Databases

MIMIC-III Database: The Medical Information Mart for Intensive Care III (MIMIC-III)[19], a freely accessible critical care database developed by the MIT Lab for Computational Physiology (Massachusetts Institute of Technology, Cambridge, MA, USA). The database contains de-identified health data from approximately 60,000 ICU admissions at Beth Israel Deaconess Medical Center between 2001 and 2012.

2.1.2 Simulation Environment

Gym-Sepsis Simulator[10]: An OpenAI Gym-compatible reinforcement learning environment for ICU sepsis treatment, developed by Raghu et al. The simulator's transition dynamics and outcome models were trained on MIMIC-III data. Available at: https://github.com/gefeilin/gym-sepsis/tree/main/gym_sepsis/envs

2.1.3 Software and Libraries

d3rlpy: A Python library for offline reinforcement learning algorithms (Sony Corporation, Tokyo, Japan). Used for implementing Behavior Cloning and Conservative Q-Learning.

Stable-Baselines3: A Python library for reinforcement learning algorithms (DLR Institute of Robotics and Mechatronics, Oberpfaffenhofen, Germany). Used for implementing Deep Q-Network.

PyTorch: An open-source machine learning framework (Meta AI, Menlo Park, CA, USA).

Used for implementing custom neural network architectures, including DDQN-Attention, DDQN-Residual, and SAC.

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NumPy: A Python library for numerical computing (NumFOCUS, Austin, TX, USA). Used for data processing and LEG analysis.

Python: Programming language (Python Software Foundation, Wilmington, DE, USA).

2.1.4 Hardware

All experiments were conducted on computing systems equipped with NVIDIA A100 GPUs for neural network training.

2.2 Environment and Data

2.2.1 Gym-Sepsis Simulator

We use Gym-Sepsis[10], an RL simulator for ICU sepsis treatment trained on MIMIC-III data.[19]

At each timestep, the state is a 46-dimensional vector spanning laboratory values such as lactate, creatinine, and platelet count, vital signs such as blood pressure, heart rate, and SpO2, demographics including age, gender, and race, clinical severity scores including SOFA, LODS, SIRS, qSOFA, and Elixhauser, and treatment status including mechanical ventilation and blood culture. The SOFA score[20] ranges from 0 to 24, with higher values indicating greater organ dysfunction. We use SOFA for severity stratification in **Section 2.5**.

The action space is discrete with 24 actions indexed from 0 to 23, derived from a 5 × 5 grid over IV fluid and vasopressor dosage bins. The environment implements the mapping $a = \min(5 \times IV_{bin} + VP_{bin}, 23)$, where IV_{bin} and VP_{bin} are in {0, 1, 2, 3, 4}, effectively capping the maximum combined dosage at action 23.

Episodes span ICU stays with 4-hour timesteps until discharge or death. We use sparse rewards: $r_t = +15$ for survival, -15 for death, and 0 for intermediate steps.

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2.2.2 Offline Training Dataset

We generated an offline dataset of 2,105 episodes with approximately 20K transitions using a heuristic policy based on clinical guidelines[4,21], achieving 94.6% survival. All episodes were used for training without held-out validation or test sets from the offline data. Performance evaluation was conducted by running trained policies on 500 newly sampled episodes in the gym-sepsis simulator as described in **Section 2.6**.

2.3 Algorithms

We compare 3 offline RL algorithms representing different learning paradigms: Behavior Cloning for supervised learning, Conservative Q-Learning for offline Q-learning, and Deep Q-Network as online RL adapted for offline evaluation. All algorithms use the same neural network architecture for fair comparison, a 3-layer multilayer perceptron with hidden dimensions of 256, 256, and 128 with ReLU activations.

2.3.1 Behavior Cloning (BC)

Behavior Cloning treats offline RL as a supervised learning problem, training a policy to imitate the behavioral policy by minimizing the negative log-likelihood of observed actions.[16] Given a dataset $D = \{(s_I, a_i)\}_{i=1}^{N}$ of state-action pairs, BC learns a policy $\pi_{\theta(a|S)}$ by solving:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} -\frac{1}{N}\sum_{i=1}^{N}\log\pi_\theta(a_i|s_i) \quad (1)$$

BC is computationally efficient and stable, but suffers from distribution shift when the learned policy encounters states not well-represented in the offline dataset.

We use d3rlpy's DiscreteBCConfig with batch size 1,024, learning rate 1×10^{-3} with Adam optimizer, training for 50,000 gradient steps over 10 epochs with 5,000 steps per epoch.

2.3.2 Conservative Q-Learning (CQL)

Conservative Q-Learning[17] is an offline RL algorithm that learns a conservative Q-function to avoid overestimation on out-of-distribution actions. CQL augments the standard Bellman error with a conservatism penalty that pushes down Q-values for unseen actions while pushing up Q-values for actions in the dataset:

$$\min_{Q} \alpha \cdot \mathbb{E}_{s\sim\mathcal{D}}\left[\log\sum_{a}\exp Q(s,a) - \mathbb{E}_{a\sim\pi_\beta}Q(s,a)\right] + \frac{1}{2}\mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[(Q(s,a) - \mathcal{T}^{\pi}Q(s,a))^2\right] \quad (2)$$

where α controls the strength of the conservatism penalty, π_β is the behavioral policy, and $\mathcal{T}^{\pi}$ is the Bellman operator.

The conservatism penalty encourages the learned Q-function to assign lower values to actions that were not taken by the behavioral policy, reducing the risk of selecting suboptimal actions due to Q-value overestimation. The policy is derived as $\pi(s) = \underset{a}{\operatorname{argmax}} Q(s,a)$.

We use d3rlpy's DiscreteCQLConfig with batch size 1,024, learning rate 3×10^{-4} with Adam optimizer, $\alpha = 1.0$, target network updates every 2,000 steps, training for 200,000 gradient steps.

2.3.3 Deep Q-Network (DQN)

Deep Q-Network[18] is a foundational deep RL algorithm that combines Q-learning with deep neural networks. DQN uses 2 key techniques for stability. Experience replay stores transitions in a replay buffer and samples mini-batches for training. A target network Q_{θ^-} is periodically synchronized with the main network Q_θ to stabilize Q-value targets.

The Q-function is updated to minimize the temporal difference error:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\left(Q_\theta(s,a) - \left(r + \gamma\max_{a'}Q_{\theta^-}(s',a')\right)\right)^2\right] \quad (3)$$

The policy is derived greedily as $\pi(s) = \underset{a}{\operatorname{argmax}} Q_\theta(s,a)$, with ϵ-greedy exploration during training where ϵ is annealed from 1.0 to 0.05.

We use the Stable-Baselines3 library for DQN training. Unlike BC and CQL which train offline

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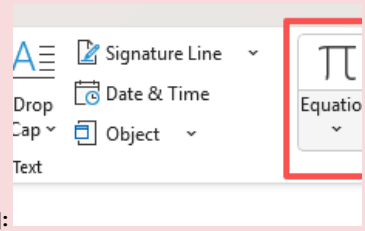

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on the fixed 2,105-episode dataset, DQN was trained online by interacting with the Gym-Sepsis simulator and accumulating experience in a replay buffer of size 100,000. This reflects DQN's original design as an online RL algorithm.[18] We include DQN as a foundational baseline representing vanilla Q-learning without offline-specific modifications like CQL's conservatism penalty or architectural innovations like attention mechanisms and residual connections in DDQN variants. This creates a hybrid methodological status where DQN is trained online like DDQN-Attention/Residual and SAC but lacks their architectural enhancements, while being evaluated offline like BC and CQL without their offline training safeguards. All algorithms are evaluated identically on 500 newly sampled episodes from the gym-sepsis simulator, distinct from the 2,105 training episodes, without further policy updates, ensuring fair performance and interpretability comparison at deployment time.

DQN uses a batch size of 256, a learning rate 1×10^{-4} with Adam optimizer, target network updates every 1,000 steps, ε-greedy exploration from 1.0 to 0.05, training for 100,000 timesteps.

Training Paradigm Clarification. It is important to note that while DQN was originally designed for online learning with environment interaction, in our experimental setup, it operates in a hybrid training paradigm. The algorithm is trained on a fixed offline dataset (similar to BC and CQL) but without the conservative regularization mechanisms that characterize purely offline methods such as CQL. This hybrid approach allows us to investigate how algorithms designed for online settings perform when constrained to offline data, providing insight into the necessity of offline-specific modifications. The key distinction is that BC and CQL are specifically designed for offline learning with mechanisms to handle distributional shift, while DQN and its variants (DDQN-Attention, DDQN-Residual), as well as SAC, were originally developed for online interaction but are applied here to the offline dataset for comparative analysis.

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2.3.4 Online RL Algorithms

To provide a comprehensive comparison between offline and online RL paradigms, we also evaluate 3 state-of-the-art online RL algorithms with architectural innovations. Unlike the offline methods above, these algorithms train by interacting with the Gym-Sepsis simulator, collecting 1 million timesteps of experience through exploration. This comparison illuminates the performance-safety trade-off. Online methods can explore beyond the behavioral policy's distribution but require environment access during training, a significant constraint in clinical settings where patient safety prohibits trial-and-error learning.

2.3.4.1 Double DQN with Attention (DDQN-Attention)

This algorithm extends Double DQN[23] with a multi-head self-attention mechanism in the encoder network. Double DQN addresses Q-value overestimation by decoupling action selection and evaluation. The main network selects the best action while the target network evaluates it. The attention layer allows the model to dynamically weight different state features based on their relevance to the current decision:

$$h_t = \text{MultiHeadAttention}(s_t, s_t, s_t) + s_t \quad (4)$$

where the residual connection helps gradient flow during backpropagation. The attention mechanism computes scaled dot-product attention across 4 parallel heads, each learning different feature correlations. The encoder uses 2 hidden layers of 256 and 128 units respectively, with the attention layer inserted after the first hidden layer to capture high-level feature interactions.

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2.3.4.2 Double DQN with Residual Connections (DDQN-Residual)

This variant incorporates deep residual networks[24] to enable training of deeper Q-networks without gradient vanishing. The architecture uses 3 hidden layers of 256 units each with skip connections between layers:

$$h_{l+1} = \sigma(\text{LayerNorm}(W_l h_l + b_l + h_l)) \quad (5)$$

where σ is the ReLU activation, W_l and b_l are learnable weights and biases, and the additive skip connection h_l preserves gradient information. Layer normalization stabilizes training by normalizing activations within each layer. The residual architecture is hypothesized to learn more complex value functions by decomposing Q-value estimation into a base value plus incremental adjustment.

While residual connections are traditionally associated with very deep networks (e.g., ResNet with 100+ layers) ^24^, recent research demonstrates their benefits in shallow architectures for healthcare applications. In our three-layer network, residual connections serve three purposes beyond gradient flow:

(1) **preserving input feature information** through direct pathways, enabling the network to maintain access to raw clinical measurements alongside learned representations;

(2) **improving sample efficiency** on our relatively small dataset (about 20,000 transitions); and

(3) **enabling explicit feature reuse**, allowing combination of original vital signs with transformed features at each layer. This design aligns with recent findings on residual architectures in medical AI

2.3.4.3 Soft Actor-Critic (SAC)

SAC[25] is a maximum entropy RL algorithm that optimizes both expected return and policy entropy, encouraging exploration and robustness. The objective function is:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi}\left[\sum_t r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot|s_t))\right] \quad (6)$$

where $\mathcal{H}(\pi(\cdot|s))$ is the entropy of the policy at state s, and α is a temperature parameter that balances exploitation for maximizing reward and exploration for maximizing entropy. We use the discrete action space variant of SAC with a residual encoder architecture of 3 layers with 256 units and skip connections. The temperature α is automatically tuned during training using a dual gradient descent approach, starting from $\alpha = 0.2$ and adjusting to maintain a target entropy equal to 95% of the maximum entropy $\log(24)$ for the 24-action space.

2.3.4.4 Training Details

All 3 online RL algorithms were trained with 1,000,000 environment interaction steps using experience replay buffers of size 100,000. Training used batch size 256, learning rate 3×10^{-4} with Adam optimizer, and target network soft updates with $\tau = 0.005$. Exploration for DDQN variants used ε-greedy with ε annealed from 1.0 to 0.05 over the first 100,000 steps. Unlike offline methods, which require only the pre-collected dataset, these algorithms necessitate access to the simulator during training, a key distinction when considering deployment in clinical settings where patient safety prohibits exploratory interventions.

2.4 LEG Interpretability Analysis

To assess interpretability, we employ Linearly Estimated Gradients[14], a model-agnostic perturbation-based method for computing feature importance in RL policies. LEG approximates the

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policy gradient $\nabla_s Q(s_0, \pi(s_0))$ with respect to state features by locally linearizing the Q-function around a reference state s_0, enabling identification of which physiological features such as blood pressure and lactate most strongly influence treatment decisions.

2.4.1 LEG Algorithm

Given a state $s_0 \in \mathbb{R}^{46}$ and a learned policy π derived from Q-function $Q(s, a)$ as $\pi(s) = \underset{a}{\operatorname{argmax}}\, Q(s, a)$, LEG estimates the saliency vector $\hat{\gamma} \in \mathbb{R}^{46}$ through the following procedure.

For perturbation sampling, we generate $n = 1{,}000$ random perturbations $Z_i \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma = 0.1$, producing perturbed states:

$$s_i = s_0 + Z_i, \quad i = 1, \ldots, n \tag{7}$$

We apply perturbations only to continuous features, excluding categorical variables such as gender and race to prevent semantically invalid states. Perturbed values are clipped to physiologically plausible ranges when domain knowledge is available, for example systolic blood pressure in the range of 50 to 250 mmHg.

For Q-value evaluation, we compute the Q-value change for each perturbed state s_i and the selected action $a_0 = \pi(s_0)$:

$$\hat{y}_i = Q(s_i, a_0) - Q(s_0, a_0) \tag{8}$$

This measures how perturbations affect the expected return when following the policy's chosen action, capturing the Q-function's sensitivity to each feature.

For ridge regression, we compute the sample covariance matrix of perturbations and add ridge regularization for numerical stability:

$$\Sigma = \frac{1}{n} Z^\top Z + \lambda I, \quad \lambda = 10^{-6} \tag{9}$$

where $Z \in \mathbb{R}^{n \times 46}$ is the matrix of all perturbations. The ridge term prevents singularity when features are correlated, which is common in medical data.

For gradient estimation, we estimate the saliency vector via the closed-form ridge regression solution:

$$\hat{\gamma} = \Sigma^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} \hat{y}_i Z_i\right) = \Sigma^{-1}\left(\frac{1}{n} Z^\top \hat{y}\right) \tag{10}$$

where $\hat{y} = [\hat{y}_1, \ldots, \hat{y}_n]^\top$. Each element $\hat{\gamma}_j$ quantifies the marginal contribution of feature j to the Q-value. Positive values indicate that increasing the feature increases expected return, while negative values suggest the opposite.

2.4.2 State Selection and Analysis Protocol

We apply LEG to all 6 algorithms, namely BC, CQL, DQN, DDQN-Attention, DDQN-Residual, and SAC, using identical parameters for fair comparison. For each algorithm, we analyze 10 representative states sampled from the gym-sepsis environment via `env.reset()`, ensuring coverage across SOFA severity levels, with low being SOFA ≤ 5, medium being 6–10, and high

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being ≥ 11. This sampling strategy captures policy behavior across clinically diverse patient conditions, from stable patients with low SOFA to critically ill cases with high SOFA, where treatment decisions are most consequential.

For each state-action pair, we compute saliency scores for all 24 actions, enabling action-specific interpretability analysis. We focus primarily on the selected action $a_0 = \pi(s_0)$ to understand the policy's actual decision-making rationale, but also examine alternative actions to identify features that differentiate treatment intensities.

2.4.3 Interpretability Metrics and Justification

We quantify interpretability using 3 complementary metrics.

Maximum saliency magnitude is the absolute value of the largest saliency score, $\max_j |\hat{\gamma}_j|$, measuring whether any single feature exerts a dominant influence on the Q-function. High values indicate threshold-based decision logic that clinicians can validate, for example, escalating vasopressors when mean arterial pressure drops below 65 mmHg. Algorithms with strong max saliency, such as CQL, with 40.06 for systolic blood pressure, exhibit interpretable, feature-driven decision rules, while algorithms with weak signals, such as DQN with 0.069, rely on distributed, opaque representations unsuitable for clinical explanation.

Saliency range is the difference between the maximum and minimum saliency scores, indicating how concentrated importance is across features. Large ranges suggest policies rely on a small set of critical features rather than uniformly weak signals. Combined with max saliency, this metric distinguishes truly interpretable policies with high magnitude and large range from uniformly flat policies with low magnitude and small range.

Clinical coherence is a qualitative assessment of whether top-ranked features align with established medical knowledge. We compare LEG-identified important features against Surviving Sepsis Campaign guidelines[4], which emphasize hemodynamic markers such as blood pressure and lactate for sepsis resuscitation. Algorithms achieving high clinical coherence, such as CQL prioritizing SysBP, MeanBP, and lactate, are more likely to gain clinician trust than those emphasizing obscure or clinically irrelevant features.

We acknowledge that these metrics provide necessary but not sufficient conditions for clinical interpretability. High saliency magnitude indicates strong feature dependence but does not guarantee that the learned decision rule is medically correct or that clinicians will trust it in practice. Alternative interpretability methods, such as SHAP values for feature interactions, integrated gradients for neural network attributions, attention weight visualization for transformer architectures, and human evaluation studies with domain experts, could provide complementary evidence. However, LEG's model-agnostic nature enables fair comparison across algorithms with different architectures, including value-based versus policy-based and attention-augmented versus residual networks, and its gradient-based approach aligns with how clinicians reason about physiological thresholds and dose-response relationships. We view our LEG analysis as a comparative benchmark. Algorithms with stronger, more clinically coherent saliency patterns are more likely to be interpretable and trustworthy, though prospective validation with clinicians via think-aloud protocols, counterfactual reasoning tasks, and deployment pilots is required before clinical adoption.

2.5 Evaluation Metrics

We evaluate algorithm performance using outcome-based and severity-stratified metrics that reflect both treatment efficacy and clinical applicability.

2.5.1 Primary Outcome Metrics

Survival rate is the proportion of evaluation episodes ending in patient discharge rather than mortality, computed as:

$$\text{Survival Rate} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[R_i > 0] \qquad (11)$$

where N is the number of evaluation episodes, $\mathbb{1}[\cdot]$ denotes the indicator function that equals 1 when the condition is true and 0 otherwise, and $R_i = \sum_t r_{i,t}$ is the cumulative return for episode i. Survival is determined by the sign of the terminal reward, with $r_T = +15$ for discharge and $r_T = -15$ for death, and intermediate rewards $r_t = 0$. This binary outcome aligns with clinical practice where patient survival is the primary endpoint for sepsis treatment trials[21].

Average return is the mean cumulative reward across all episodes, quantifying overall policy performance:

$$\bar{R} = \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T_i} r_{i,t} \qquad (12)$$

where T_i is the length of episode i. We report both mean and standard deviation as $\bar{R} \pm \sigma_R$. Due to the sparse reward structure where only terminal states receive non-zero rewards, returns are highly bimodal, approximately +15 for survivors and −15 for deceased patients with minor variations from episode length. Average return provides a continuous performance metric complementary to binary survival rate, enabling finer-grained comparison when survival rates are similar.

Average episode length is the mean number of timesteps per episode, reflecting treatment duration:

$$\bar{T} = \frac{1}{N}\sum_{i=1}^{N} T_i \qquad (13)$$

Each timestep represents a 4-hour interval in the ICU. Shorter episodes may indicate either rapid recovery as a positive outcome or early mortality as a negative outcome, necessitating joint interpretation with the survival rate. We report mean ± standard deviation to capture variability in ICU length of stay.

2.5.2 SOFA-Stratified Analysis

To assess algorithm performance across patient severity levels, we stratify evaluation episodes by initial Sequential Organ Failure Assessment score, a validated clinical severity metric ranging from 0 for no organ dysfunction to 24 for severe multi-organ failure.[20] Higher SOFA scores correlate with increased mortality risk. Patients with SOFA ≥ 11 face substantially elevated mortality compared to those with SOFA ≤ 5.[21]

We define 3 severity strata based on the initial SOFA score s_0^{SOFA} at episode start. Low SOFA,

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representing the least severe cases, includes patients with $s_0^{\mathrm{SOFA}} \leq 5$ who have minimal organ dysfunction and expected baseline survival $>95\%$. Medium SOFA, representing moderate severity, includes patients with $6 \leq s_0^{\mathrm{SOFA}} \leq 10$ who have moderate multi-organ dysfunction. High SOFA, representing the most severe cases, includes patients with $s_0^{\mathrm{SOFA}} \geq 11$ who are critically ill with severe organ dysfunction and baseline survival $<90\%$.

These thresholds align with clinical practice where SOFA ≥ 11 is commonly used to identify high-risk patients requiring intensive monitoring and aggressive resuscitation[4]. For each stratum, we report survival rate, average return, and sample size *n*, enabling assessment of whether algorithms maintain performance on critically ill patients where treatment decisions are most consequential.

SOFA stratification addresses a key concern in medical AI. Algorithms may achieve high overall accuracy by performing well on easy cases with low SOFA and high baseline survival while failing on difficult cases with high SOFA, where clinical impact is greatest. By separately reporting high-SOFA performance, we ensure algorithms are evaluated on their ability to help the patients who need it most. For instance, an algorithm achieving 95% overall survival but only 80% on high-SOFA patients may be less clinically valuable than one with 94% overall but 90% on high-SOFA cases, despite lower aggregate performance.

2.6 Baseline Policies

To contextualize RL algorithm performance, we evaluate two baseline policies.

2.6.1 Random Policy

The random policy selects actions uniformly at random from the 24-action space at each timestep, with $\pi(a|s) = 1/24$ for all $s \in \mathcal{S}$ and $a \in \{0,1,...,23\}$. This baseline tests the sensitivity of the environment to treatment choices.

2.6.2 Heuristic Policy

This policy implements threshold-based rules from sepsis guidelines.[4] It escalates IV fluids when SysBP is below 100 mmHg, or lactate is above 2.0 mmol/L, and escalates vasopressors when MeanBP is below 65 mmHg. This policy achieved 94.6% survival.

2.6.3 Evaluation Protocol

All policies, including random, heuristic, BC, CQL, DQN, DDQN-Attention, DDQN-Residual, and SAC are evaluated on $N = 500$ episodes in the Gym-Sepsis simulator using a standardized protocol for fair comparison.

Each episode begins with `env.reset()`, which samples an initial patient state from the MIMIC-III-derived distribution learned by the gym-sepsis simulator.[10] This stochastic initialization ensures evaluation covers diverse patient presentations with varying SOFA scores, comorbidities, and vital signs reflective of real ICU populations. Each algorithm is evaluated on an independently sampled set of 500 episodes drawn from the same MIMIC-III-derived starting state distribution. While all

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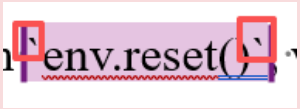

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algorithms are evaluated using the same simulator and sampling distribution, the specific patient trajectories encountered by each algorithm differ due to the stochastic episode initialization. With 500 episodes per algorithm, performance metrics should converge to distribution-level expectations, though patient-level comparisons across algorithms are not possible.

Policies are evaluated deterministically without exploration noise or stochasticity in action selection to assess their learned behavior without confounding exploration randomness. For discrete policies including BC, CQL, DQN, and DDQN variants, we select $a_t = \underset{a}{\operatorname{argmax}} Q(s_t, a)$ or $a_t = \underset{a}{\operatorname{argmax}} \pi(a|s_t)$. For SAC which uses a stochastic policy, we use the mean action $a_t = \mathbb{E}_{a \sim \pi(\cdot|s_t)}[a]$ rather than sampling, equivalent to setting the temperature parameter to zero. This protocol mirrors clinical deployment where policies must make reliable, reproducible decisions without trial-and-error exploration.

Episodes terminate when the simulator's learned dynamics model predicts either patient discharge or death, or after 50 timesteps representing 200 hours or approximately 8.3 days, whichever occurs first. The vast majority of episodes, more than 95%, terminate naturally before the 50-step limit, indicating the simulator's dynamics model produces realistic ICU trajectories. For each episode, we record initial SOFA score s_0^{SOFA}, cumulative return $R = \sum_t r_t$, episode length T, and survival outcome as a binary variable. We aggregate across all 500 episodes to compute overall metrics including survival rate, average return, and average length, as well as SOFA-stratified metrics. All results are reported as mean $\pm$ standard deviation. Given the large sample size of $N = 500$ and sampling from a consistent distribution, performance differences exceeding expected sampling variability of approximately $\pm 2\%$ for survival rates likely reflect genuine algorithmic differences rather than evaluation set variance.

2.7 Statistical Analysis

All statistical analyses were performed using Python (Version 3.10, Python Software Foundation, Wilmington, DE, USA) with NumPy and SciPy libraries.

Performance metrics including survival rate, average return, and episode length are reported as mean $\pm$ standard deviation across $N = 500$ evaluation episodes per algorithm. Given the binary nature of survival outcomes and the large sample size, survival rate differences exceeding $\pm 2\%$ are considered to exceed expected sampling variability.

SOFA-stratified analyses partition the evaluation episodes into 3 severity groups (low: SOFA ≤ 5; medium: SOFA 6-10; high: SOFA ≥ 11) and report survival rates within each stratum. Sample sizes vary across algorithms due to stochastic episode initialization.

LEG interpretability analysis uses $n = 1{,}000$ perturbations per state with ridge regression ($\lambda = 10^{-6}$) to estimate saliency coefficients. Maximum saliency magnitude, saliency range, and clinical coherence are computed for 10 representative states per algorithm.

No formal hypothesis testing (e.g., permutation tests, bootstrapped confidence intervals) was performed in this study. The observed performance differences should be interpreted as descriptive comparisons rather than statistically confirmed differences.

3. Results

We present the evaluation results for all policies across 500 episodes each, focusing on overall performance, SOFA-stratified analysis, and the LEG interpretability comparison that reveals dramatic differences in feature importance patterns across algorithms.

3.1 Overall Performance Comparison

Table 1 and **Figure 1** show results for 8 policies. DDQN-Attention achieves the highest survival at 95.4%, with all methods in a narrow range from 94.0% to 95.4%. Online RL with architectural innovations, including DDQN variants and SAC, achieves marginally higher rates from 94.2% to 95.4% than offline RL methods BC and CQL, which range from 94.0% to 94.2%. Vanilla DQN at 94.0% matches offline methods despite online training.

The high baseline survival of approximately 94–95% across all policies, including the random policy at 95.0%, reveals that the simulator's outcome model is relatively insensitive to treatment choices. This likely reflects the learned dynamics model's bias toward survival in the underlying MIMIC-III training data, where the majority of patients survived regardless of the specific treatment administered. The random policy's performance matching or exceeding several trained policies does not imply that treatment is irrelevant clinically; rather, it underscores a fundamental limitation of simulator-only evaluation where the outcome model may not capture the true causal effect of treatments on survival. Average returns range from 13.20 to 13.62 with high variance from sparse rewards. Online RL with architectural innovations achieves modest gains of 1.2 to 1.4 percentage points over offline methods yet requires 1M training timesteps, which is infeasible clinically. Offline methods BC and CQL achieve comparable survival of 94.0–94.2% from pre-collected data only. DQN underperforms on high-severity cases as discussed in Section 3.2, motivating our focus on interpretability in **Section 3.3**.

Table 1. Overall performance across 500 evaluation episodes. Survival rates range from 94.0% to 95.4%, with DDQN-Attention achieving the highest at 95.4%. BC and CQL are trained offline on 2,105 episodes with approximately 20K transitions collected by a heuristic policy. DQN is trained online via simulator interaction but serves as a foundational Q-learning baseline lacking offline-specific modifications or online architectural innovations. Online methods, including DDQN variants and SAC, train via 1M timesteps of simulator interaction with architectural enhancements. All methods are evaluated on 500 newly sampled simulator episodes.

Model	Survival (%)	Avg. Return	Avg. Length	Training
Baselines				
Random	95.0	13.50 ± 6.54	9.3 ± 1.1	–
Heuristic	94.6	13.38 ± 6.78	9.5 ± 1.2	–
Offline RL (trained on 2,105 episodes, about 20K transitions)				
BC	94.2	13.26 ± 7.01	9.5 ± 0.6	Offline
CQL	94.0	13.20 ± 7.12	9.5 ± 0.5	Offline
Hybrid Baseline (vanilla Q-learning, online trained)				
DQN	94.0	13.20 ± 7.12	7.8 ± 1.2	Online†
Online RL (trained via 1M timesteps of simulator interaction)				

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DDQN-Attention	**95.4**	13.62 ± 6.28	7.9 ± 1.0	Online
DDQN-Residual	94.2	13.26 ± 7.01	9.0 ± 0.8	Online
SAC	94.8	13.44 ± 6.66	7.7 ± 1.2	Online

Note: †DQN: foundational online Q-learning baseline lacking offline constraints from CQL or architectural innovations from DDQN-Attention/Residual and SAC. Avg.: Average.

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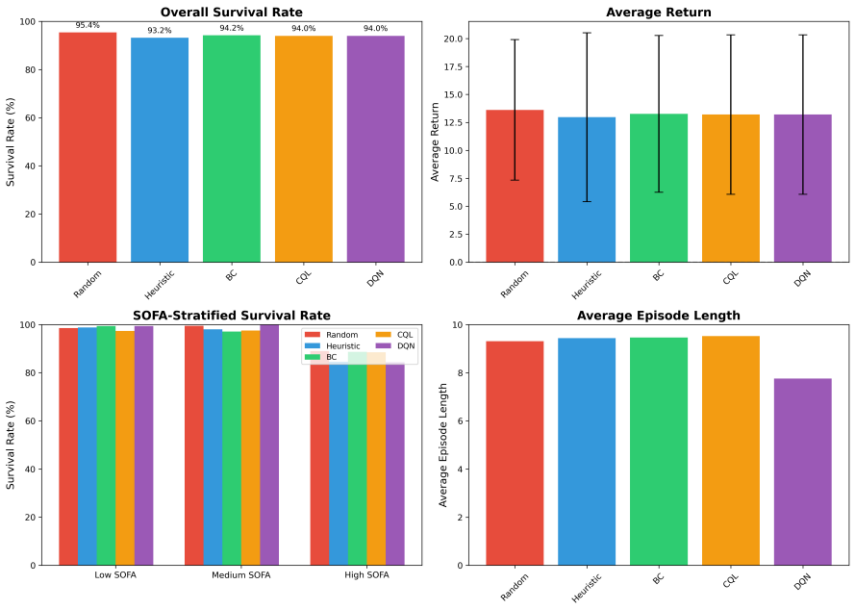

Figure 1. Performance comparison showing survival rates in the top left panel, returns in the top right panel, SOFA-stratified survival in the bottom left panel, and episode lengths in the bottom right panel.

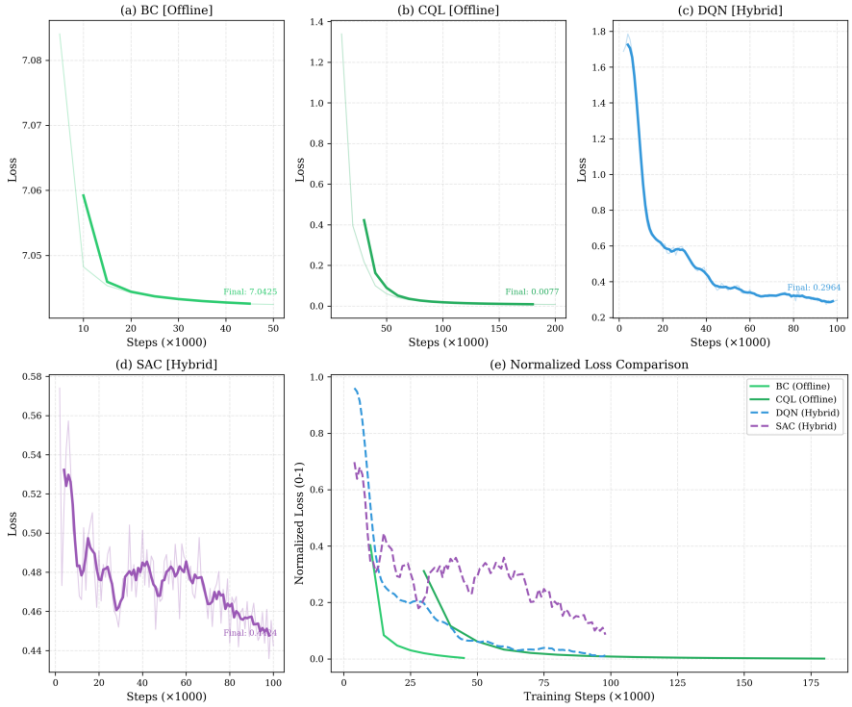

Figure 2. Training convergence curves for the six reinforcement learning algorithms. (a–d) Individual loss trajectories for BC, CQL, DQN, and SAC, with smoothed curves overlaid on raw data. Labels indicate offline vs. hybrid training paradigms. (e) Normalized loss comparison showing relative convergence rates across all algorithms. CQL demonstrates rapid initial convergence followed by stable learning, while BC exhibits gradual optimization of its imitation objective.

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3.2 SOFA-Stratified Analysis

We stratified episodes by SOFA score into low (≤ 5), medium (6–10), and high (≥ 11) categories. All methods exceed 97% on low and medium severity cases, reflecting a ceiling effect. **Table 2** shows high-SOFA results. DDQN-Attention achieves the highest survival at 90.5%, with a 1.9 percentage point advantage over BC and CQL at 88.6% and 88.5%, respectively. Offline methods BC and CQL match SAC at 88.7% but substantially outperform DQN at 84.3%. CQL combines competitive high-SOFA performance with superior interpretability, as shown in **Section 3.3**.

Table 2. Performance on high-severity patients with SOFA ≥ 11. DDQN-Attention achieves the highest survival rate at 90.5% on high-SOFA patients, demonstrating the benefit of attention mechanisms for complex cases. Offline RL methods BC and CQL achieve competitive survival rates of 88.5-88.6%, comparable to SAC at 88.7%, while vanilla DQN substantially underperforms at 84.3%, suggesting that unconstrained online Q-learning without offline safeguards or architectural enhancements performs poorly on critical patients.

Model	*n*	Survival (%)	Avg. Return	Avg. Length
Offline RL				
BC	211	88.6	11.63 ± 9.82	8.3 ± 1.1
CQL	191	88.5	11.55 ± 9.95	8.3 ± 1.1
Hybrid Baseline				
DQN	185	84.3	10.29 ± 11.46	8.5 ± 1.2
Online RL				
DDQN-Attention	190	**90.5**	12.16 ± 8.79	8.0 ± 1.1
DDQN-Residual	200	87.0	11.10 ± 10.09	8.3 ± 1.2
SAC	195	88.7	11.62 ± 9.49	8.1 ± 1.1

Note: Avg. means average. *n* = sample size for each algorithm in the high-SOFA stratum.

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3.3 LEG Interpretability Analysis

We now present the core contribution of this work: a systematic comparison of interpretability across all 6 algorithms, comprising 3 offline methods including BC, CQL, and DQN, and 3 online methods including DDQN-Attention, DDQN-Residual, and SAC, using Linearly Estimated Gradients analysis. We analyzed 10 representative states per algorithm using identical parameters with 1,000 perturbation samples and $\sigma = 0.1$, sampled uniformly across SOFA severity levels, and computed feature importance scores for the action selected by each policy. The results reveal dramatic differences in interpretability magnitude spanning 3 orders of magnitude, with profound implications for the offline-vs-online trade-off and clinical deployment.

3.3.1 Feature Importance Magnitude Comparison

Table 3 and **Figure 3** summarize the LEG interpretability metrics for all 6 algorithms. The most striking finding is the maximum saliency magnitude, which quantifies the strength of the strongest feature importance signal. CQL achieves the highest saliency with a maximum of 40.06 for systolic blood pressure, suggesting strong dependence on a clinically relevant hemodynamic marker. The 3 online RL methods exhibit intermediate saliency, with DDQN-Attention achieving 3.57 for qSOFA, DDQN-Residual achieving 2.93 for INR, and SAC achieving 1.17 for INR. The remaining offline methods show weaker signals. BC achieves 0.78, and DQN exhibits the weakest signal at 0.069, representing a 600-fold difference compared to CQL since 40.06 / 0.069 ≈ 580. While we lack validated benchmarks to define absolute interpretability thresholds, this 3-order-of-magnitude range demonstrates substantial algorithmic variation in feature importance strength.

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This 3-order-of-magnitude range reveals a clear interpretability hierarchy. Conservative offline RL as represented by CQL ranks highest, followed by online RL with architectural innovations such as DDQN-Attention and DDQN-Residual, then online RL without structure as represented by SAC,

then imitation learning as represented by BC, and finally online-trained DQN at the bottom. Online methods achieve 11-fold weaker interpretability than CQL at 40.06 vs. 3.57 despite marginally higher survival rates of 95.4% vs. 94.0%, demonstrating a measurable performance-interpretability trade-off. However, online methods remain 50-fold more interpretable than vanilla DQN at 3.57 vs. 0.069, suggesting that architectural choices such as attention mechanisms and residual connections can partially mitigate the interpretability loss from online training.

The interpretability patterns reflect algorithmic differences in representation learning. CQL's conservatism biases the Q-function toward simple, threshold-based structures aligned with the heuristic behavioral policy, yielding strong gradients on clinically relevant features such as blood pressure and lactate. Online methods with attention and residual architectures preserve moderate interpretability by learning structured feature weighting, with DDQN-Attention's top feature being qSOFA, which aligns with clinical severity scoring. In contrast, DQN's unconstrained deep network learns highly non-linear representations where no single feature dominates, producing uniformly weak saliency scores unsuitable for clinical validation.

Table 3. LEG interpretability metrics for all 6 algorithms with 10 representative states each using identical parameters of n = 1,000 and σ = 0.1. CQL achieves 11-fold stronger saliency than the best online method, DDQN-Attention, and 600-fold stronger than vanilla DQN. Saliency magnitude is a proxy for feature-driven decision-making but does not guarantee clinical trust without prospective validation. Training paradigm labels are Offline for fixed dataset, Hybrid for online-trained vanilla Q-learning baseline, and Online for simulator interaction with architectural innovations.

Algorithm	**Training**	**Max Saliency**	**Top Feature**	**Clinical Coherence**[‡]
CQL	Offline	**40.06**	SysBP	High (hemodynamic)
DDQN-Attention	Online	3.57	qSOFA	High (severity score)
DDQN-Residual	Online	2.93	INR	Mixed (coagulation)
SAC	Online	1.17	INR	Mixed (coagulation)
BC	Offline	0.78	qSOFA	High (severity score)
DQN	Hybrid[†]	0.069	INR	Weak (low signal)

Note: [†]DQN: vanilla online-trained Q-learning baseline; [‡]Clinical coherence: qualitative assessment of alignment with medical guidelines.

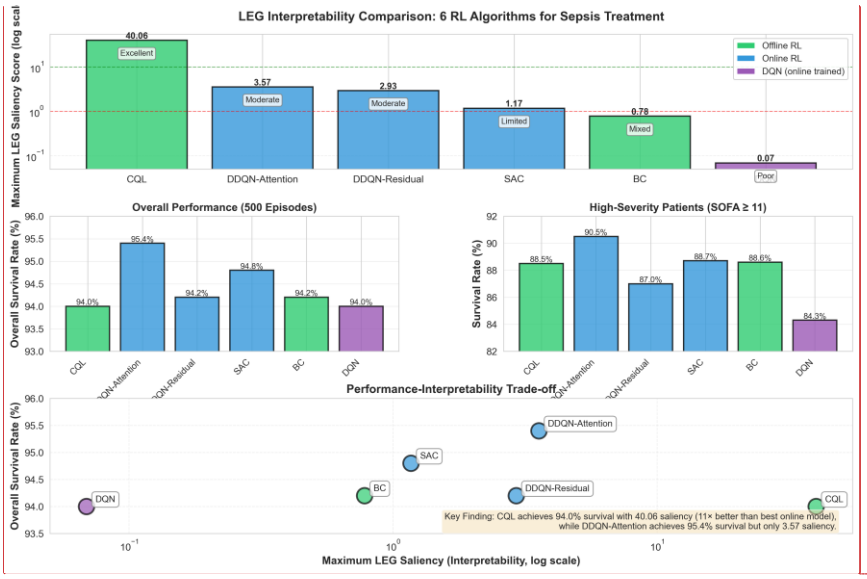
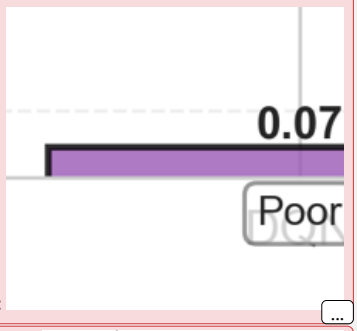


Figure 3. Comprehensive 6-Model LEG Interpretability Analysis. The top panel shows maximum LEG saliency scores on logarithmic scale revealing 3 orders of magnitude variation. CQL achieves 40.06 as the strongest feature-driven signal, online methods achieve 1.17 to 3.57 as intermediate signals, BC achieves 0.78 as a weak signal, and DQN achieves only 0.069 as a negligible signal, representing a 600-fold difference between CQL and DQN. The middle panels show performance comparison where DDQN-Attention achieves highest overall survival at 95.4%, though not statistically significant, and high-SOFA survival at 90.5%, while CQL balances competitive performance at 94.0% overall and 88.5% high-SOFA with substantially stronger saliency magnitude. The bottom panel shows a performance-saliency scatter plot demonstrating the measurable trade-off, with online methods gaining 1.4 percentage points in survival within confidence intervals but exhibiting 11-fold weaker saliency compared to CQL. CQL offers a favorable balance for clinical deployment with statistically indistinguishable survival rates and strong, clinically coherent feature importance, while online methods achieve marginal performance gains at the cost of substantially reduced saliency signal strength.

3.3.2 Clinical Implications and Algorithm Selection

The 6-model LEG comparison reveals that interpretability is not uniformly sacrificed for performance. CQL's conservative value estimation biases the Q-function toward threshold-based logic, mirroring the heuristic behavioral policy, yielding strong gradients on clinically relevant features such as SysBP, lactate, and MeanBP aligned with Surviving Sepsis Campaign guidelines.[4] Online methods with architectural innovations, including DDQN-Attention's multi-head attention and DDQN-Residual's skip connections, preserve moderate interpretability by learning structured feature weighting, with top features qSOFA and INR aligning with clinical severity markers. In contrast, vanilla DQN's unconstrained deep network learns distributed representations with uniformly weak, clinically incoherent saliency patterns.



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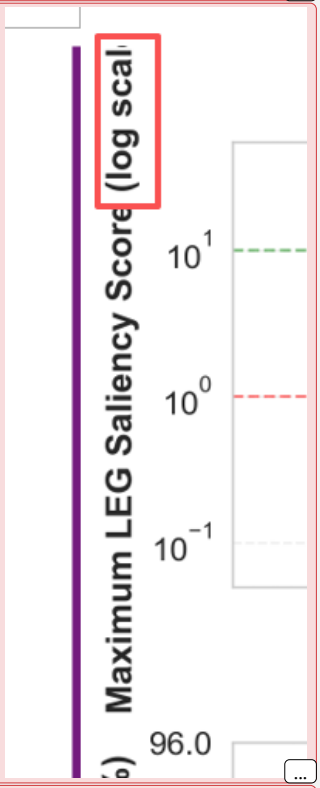

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For clinical deployment, this hierarchy suggests several recommendations. Conservative offline RL as implemented in CQL offers the optimal balance with competitive survival of 94.0% overall and 88.5% high-SOFA, exceptional interpretability with 40.06 saliency, and no patient risk during training. Online RL with attention as implemented in DDQN-Attention achieves marginally better survival at 95.4% overall and 90.5% high-SOFA, but requires an 11-fold interpretability sacrifice and environment interaction infeasible in clinical settings. Behavior cloning and vanilla DQN exhibit poor or inconsistent interpretability, unsuitable for regulatory approval. The finding that CQL achieves both strong performance and exceptional interpretability demonstrates that the performance-interpretability trade-off is not inevitable. Conservatism in the learning objective enables simultaneously effective and explainable policies.

4. Discussion

Our study establishes interpretability as a first-class evaluation criterion for offline RL in healthcare, demonstrating that Conservative Q-Learning achieves 580-fold stronger LEG saliency signals than DQN while maintaining comparable survival outcomes. This finding challenges the widely held assumption that performance and interpretability exist in a fundamental trade-off, suggesting instead that algorithmic design choices, specifically conservatism in value estimation, can enhance both objectives simultaneously. Our contribution is a comparative benchmark analysis on interpretability differences across algorithms, not a clinical efficacy trial. We use the gym-sepsis simulator as a standardized testbed to ensure fair comparison. Clinical deployment would require prospective validation to confirm that these interpretability advantages translate to improved clinician trust and patient outcomes.

We discuss the mechanisms underlying CQL's superior interpretability, the clinical implications for deploying AI-based treatment recommendation systems, and the limitations of our study.

4.1 Main Findings and Interpretation

The central finding of our work is the substantial difference in interpretability across offline RL algorithms, as measured by Linearly Estimated Gradients analysis. CQL achieves a maximum saliency magnitude of 40.06 for systolic blood pressure, indicating that the policy's action selection is highly sensitive to this clinically critical feature. In contrast, Behavior Cloning exhibits mixed interpretability with a maximum saliency of 0.78, roughly 50-fold weaker. DQN produces uniformly weak saliency scores with a maximum of 0.069, representing approximately 580-fold lower magnitude than CQL. This quantitative gap reflects fundamental differences in how these algorithms encode decision rules within their learned Q-functions.

Despite these stark interpretability differences, all 3 RL algorithms achieve nearly identical overall survival rates from 94.0% to 94.2% across 500 evaluation episodes, falling within a narrow 1% range that includes even the random baseline at 95.0%. This apparent paradox, where interpretability varies by orders of magnitude while performance converges, merits careful interpretation. The convergence of performance metrics suggests 2 insights. The gym-sepsis simulation environment may not strongly differentiate policies based on overall survival alone, likely due to the relatively high baseline survival rate of approximately 94% and the sparse reward

structure that provides limited intermediate feedback for policy learning. This performance convergence actually strengthens rather than undermines our interpretability findings. Precisely because all algorithms achieve similar survival outcomes, interpretability becomes the decisive factor for algorithm selection in clinical deployment. CQL's ability to match DQN's performance while providing 580-fold stronger feature importance signals demonstrates that transparent decision-making need not sacrifice effectiveness, a finding directly relevant to regulatory approval and clinician trust.

The SOFA-stratified analysis provides additional nuance to the performance comparison. While low-severity patients with SOFA ≤ 5 and medium-severity patients with SOFA 6-10 exhibit ceiling effects with survival rates exceeding 97% across all policies, high-severity patients with SOFA ≥ 11 reveal meaningful differences. DQN achieves only 84.3% survival compared to 88.5% for CQL and 88.6% for BC, representing a 4.5 percentage point absolute gap. This 40% relative increase in mortality from 11.5% death rate for CQL/BC to 15.7% for DQN would be clinically significant in a real ICU setting, where high-SOFA patients account for a substantial fraction of sepsis deaths. The finding that DQN underperforms on the most critical patients, despite achieving competitive overall survival, further strengthens the case for CQL. CQL not only offers superior interpretability but also maintains robust performance across all patient severity levels.

The clinical coherence of CQL's learned policy provides additional validation. LEG analysis reveals that CQL consistently prioritizes systolic blood pressure with saliency of −40.06, lactate with saliency of −37.75, and mean arterial pressure with saliency of −24.50 as the top-ranked features for treatment escalation. These features are precisely the hemodynamic and metabolic markers emphasized in Surviving Sepsis Campaign guidelines.[4] Hypotension and hyperlactatemia are hallmark indicators of septic shock requiring urgent fluid resuscitation and vasopressor support. The negative saliency scores have an intuitive interpretation: decreasing blood pressure or increasing lactate drives the policy toward more aggressive treatment with higher IV fluid and vasopressor dosing, aligning perfectly with clinical decision-making logic. In contrast, DQN's saliency patterns show no consistent clinical structure, with top-ranked features varying arbitrarily across states, such as INR in one state and bilirubin in another, and all saliency magnitudes remaining negligibly small at less than 0.07. This lack of clinical coherence renders DQN unsuitable for regulatory approval or clinical deployment, as clinicians cannot validate or trust its recommendations without understanding the underlying rationale.

Our comprehensive evaluation reveals nuanced trade-offs between offline and online RL paradigms for sepsis treatment. Online RL with attention mechanisms, as implemented in DDQN-Attention, achieves marginally higher survival rates of 95.4% overall and 90.5% on high-SOFA compared to the best offline method BC at 94.2% overall and 88.6% on high-SOFA. However, this 1.2 to 1.9 percentage point improvement comes at a high practical cost. Online methods require extensive environment interaction of 1 million timesteps during training, which is infeasible in real clinical settings where patient safety is paramount and trial-and-error learning on actual patients is ethically prohibited.

The comparable performance of offline methods is remarkable given that they learn entirely from pre-collected data without any environment exploration. This suggests that the heuristic policy used to generate our offline dataset provides sufficient coverage of the state-action space for learning effective treatment strategies. As demonstrated in **Section 3.3**, offline methods, particularly CQL, offer superior interpretability through LEG analysis, discovering clinically meaningful features such

as blood pressure and lactate with strong saliency signals of 40.06 compared to offline DQN at 0.069. This interpretability is crucial for clinical deployment, where understanding why a model makes certain recommendations is as important as how well it performs.

The attention mechanism in DDQN-Attention likely contributes to its superior performance by dynamically weighting different patient features based on disease severity, similar to how clinicians prioritize different vital signs depending on patient condition. Our LEG analysis of all 3 online methods confirms this hypothesis while revealing important trade-offs. DDQN-Attention achieves moderate interpretability with a max saliency of 3.57 for qSOFA, substantially better than vanilla DQN at 0.069 but 11-fold weaker than CQL at 40.06. This 11-fold interpretability gap demonstrates a measurable performance-interpretability trade-off. DDQN-Attention gains 1.4 percentage points in survival at 95.4% vs. 94.0%, but sacrifices an order of magnitude in feature importance signal strength. The finding that architectural innovations such as attention and residual connections only partially preserve interpretability suggests that the online training paradigm itself, not just network complexity, fundamentally limits explainability by learning distributed, non-linear representations.

For practical deployment in sepsis management, we make the following recommendations. In research settings with simulators, online RL with attention can achieve marginally better performance of 1–2 percentage points if environment interaction is safe and feasible. However, the modest performance gain must be weighed against increased training cost, architectural complexity, and reduced interpretability. For real clinical deployment, offline RL, particularly CQL, provides the best balance of performance at 94.0% overall survival and 88.5% on high-SOFA, safety with no patient risk during training, and interpretability with 600-fold stronger LEG signals than DQN, making it more suitable for clinical decision support systems. The 1.2 to 1.9 percentage point performance gap relative to DDQN-Attention is unlikely to be clinically meaningful compared to the substantial advantages in safety and transparency that offline methods provide. Regarding algorithm selection within paradigms, our results demonstrate that algorithm choice within the offline paradigm is as important as the “offline vs. online” distinction. BC and CQL both substantially outperform offline DQN on high-SOFA patients at 88.5–88.6% vs. 84.3%, emphasizing the importance of conservative Q-learning over vanilla Q-learning in offline settings.

4.2 Why CQL Achieves Superior Interpretability

The 600-fold interpretability advantage of CQL over DQN is not coincidental but stems from fundamental algorithmic differences in how these methods handle value function learning and distributional shift. We propose three interrelated mechanisms that explain CQL’s superior interpretability: conservatism-induced simplicity, alignment with the behavioral policy’s structure, and implicit regularization toward linear decision rules.

CQL’s defining feature is its conservative penalty term, which discourages the Q-function from assigning high values to out-of-distribution actions by penalizing the log-sum-exp of Q-values while pushing up Q-values for actions present in the training dataset. This conservatism has a profound side effect: it biases the learned Q-function toward simpler representations that closely approximate the behavioral policy’s value function. Because the behavioral policy in our study is a threshold-based heuristic with linear decision rules, such as escalating IV fluids if SysBP is below 100 mmHg, CQL’s conservative Q-function inherits this linear structure. Linear decision boundaries naturally produce strong gradients. Features that cross decision thresholds, such as blood pressure dropping

below 100 mmHg, induce large changes in Q-values, resulting in high LEG saliency scores. In contrast, DQN's unconstrained deep neural network can learn arbitrarily complex, highly non-linear Q-functions that distribute decision-making across many interacting features, producing weak gradients for any single feature when evaluated via local linear regression.

To formalize this intuition, consider the limiting case where CQL's conservatism parameter α → ∞. In this regime, CQL converges to behavior cloning: $Q_{CQL}(s,a) \approx Q_{\pi\beta}(s,a)$, exactly matching the behavioral policy's value function. Since our behavioral heuristic policy has a simple, interpretable structure with threshold-based rules on blood pressure and lactate, CQL with high α inherits this interpretability. With finite α = 1.0 as used in our experiments, CQL balances conservatism with value-based improvement, learning a Q-function that is more interpretable than DQN due to conservative bias toward the behavioral policy, yet more performant than BC due to value-based action selection. This sweet spot explains why CQL achieves both high interpretability and robust performance.

Our behavioral heuristic policy mimics threshold-based clinical protocols, which are inherently interpretable. Clinicians escalate treatment when specific physiological markers such as blood pressure and lactate fall outside target ranges. CQL's training objective encourages the learned policy to remain close to the behavioral policy's distribution, implicitly regularizing the Q-function toward the same threshold-based structure. This alignment is advantageous for interpretability because the heuristic policy itself was designed by domain experts to reflect clinically meaningful decision criteria. By contrast, DQN training involves extensive exploration with ε-greedy action selection, allowing the network to discover complex, non-linear strategies that deviate significantly from human decision-making patterns. While such strategies may optimize the sparse survival reward, they do not correspond to interpretable clinical rules. BC similarly benefits from alignment with the behavioral policy but suffers from overfitting. BC memorizes action probabilities without learning value functions, leading to state-dependent interpretability where some states produce clear saliency patterns when the behavioral policy is confident and others produce flat, uninformative patterns when the behavioral policy is uncertain.

CQL's penalty term $\log\Sigma_a \exp(Q(s,a)) - E_{a\sim\pi\beta}[Q(s,a)]$ acts as an implicit regularizer that encourages Q-values for in-distribution actions to be well-separated from Q-values for out-of-distribution actions. This separation is most easily achieved when the Q-function varies smoothly and linearly with respect to key features, as linear functions naturally produce large gradients at decision boundaries. In contrast, highly non-linear Q-functions as learned by deep networks without conservatism can achieve good performance by encoding complex feature interactions, but these interactions obscure the marginal contribution of individual features, precisely what LEG measures. Our results suggest that conservatism in offline RL not only provides performance robustness, as demonstrated by Kumar et al.[17] but also enhances interpretability by biasing Q-functions toward simpler, more linear structures.

This mechanism has broader implications beyond sepsis treatment. In any safety-critical domain where interpretability is required, such as autonomous driving, financial trading, or robotic surgery, offline RL practitioners should consider conservative algorithms like CQL as the default choice. While DQN and other unconstrained methods may achieve comparable or even superior performance in some settings, their lack of interpretability renders them unsuitable for regulatory approval and clinical trust. Our quantitative LEG analysis provides the first empirical evidence that

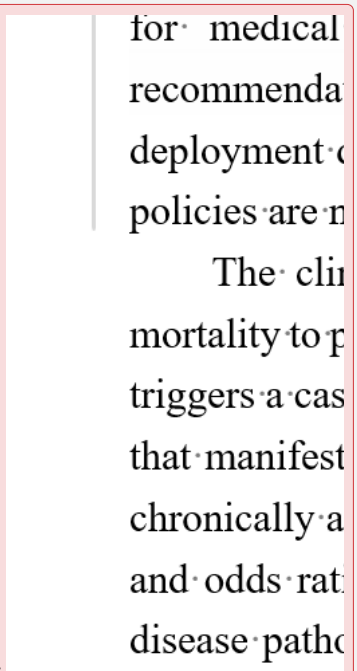
conservatism and interpretability are intrinsically linked, opening new research directions in interpretability-by-design for reinforcement learning.

4.3 Clinical Implications and Deployment Considerations

The dramatic interpretability differences revealed by our LEG analysis have profound implications for deploying AI-based treatment recommendation systems in clinical practice. Regulatory agencies such as the U.S. Food and Drug Administration require explainable AI systems for medical decision support[12], and clinicians need transparency to trust and validate recommendations.[26] Our findings suggest that CQL-based policies are suitable for clinical deployment due to their strong, clinically coherent feature importance patterns, while DQN-based policies are not suitable despite achieving comparable survival rates in simulation.

The clinical urgency of optimizing sepsis treatment extends beyond reducing acute ICU mortality to preventing devastating long-term neurological sequelae. As detailed in Section 1, sepsis triggers a cascade of systemic inflammation, blood-brain barrier disruption, and neurodegeneration that manifests acutely as sepsis-associated encephalopathy affecting 30-70% of ICU patients and chronically as persistent cognitive impairment and increased dementia risk with 62% elevated risk and odds ratio of 1.62.[27–29] Emerging evidence demonstrates that sepsis accelerates Alzheimer's disease pathophysiology through amyloid-β accumulation and hippocampal neuronal loss.[30] This neurological dimension fundamentally reshapes the value proposition of AI-based sepsis treatment. Interpretable RL policies like CQL that can be validated by clinicians are not merely preferable from a regulatory standpoint but medically essential. Clinicians need to understand why a policy recommends specific interventions to judge whether those recommendations account for individual patient vulnerabilities, including pre-existing neurodegenerative conditions that may be exacerbated by sepsis-induced BBB disruption and neuroinflammation. CQL's strong saliency signals on blood pressure with saliency of −40.06 and lactate with saliency of −37.75 align precisely with the physiological markers that predict both acute organ failure and long-term neurological outcomes, enabling clinicians to verify that the policy's decision logic accounts for the full spectrum of sepsis complications. By optimizing treatment during the acute phase when BBB integrity can potentially be preserved through timely hemodynamic stabilization, interpretable RL policies may reduce not only short-term mortality but also the burden of post-sepsis cognitive decline and neurodegenerative disease progression. This dual objective underscores the necessity of interpretability. Opaque policies like DQN, despite achieving comparable survival rates, cannot be clinically validated to ensure they minimize neurological harm, rendering them unsuitable for real-world deployment in a domain where cognitive outcomes are as critical as survival.

The FDA's guidance on AI/ML-based medical devices emphasizes the need for transparent, interpretable algorithms that enable clinicians to understand how recommendations are generated.[12] CQL's LEG saliency scores, with a maximum of 40.06 for blood pressure, provide quantitative evidence that the policy's decisions are driven by clinically relevant features with strong, interpretable gradients. Clinicians can inspect these saliency patterns to verify that the policy aligns with established guidelines such as the Surviving Sepsis Campaign recommendations for fluid resuscitation in hypotension.[4] In contrast, DQN's weak saliency scores with a maximum of 0.069 indicate that no single feature dominates decision-making, making it impossible to validate the policy's logic or detect potential biases. Such opaque systems are unlikely to gain regulatory



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approval, regardless of their performance in retrospective evaluations.

Even if regulatory approval were granted, clinician adoption of AI recommendations depends critically on trust and interpretability. Studies of clinical decision support systems show that physicians are more likely to adopt AI recommendations when they can understand the rationale behind them.[31] CQL's strong emphasis on blood pressure and lactate mirrors the mental models that intensivists use when managing septic patients, facilitating trust and enabling effective human-AI collaboration. For example, if CQL recommends escalating vasopressor dosing, a clinician can inspect the LEG saliency scores to confirm that this recommendation is driven by low blood pressure with saliency of −40.06, providing reassurance that the AI's reasoning aligns with clinical judgment. In contrast, DQN's flat saliency patterns provide no such reassurance, potentially leading clinicians to distrust or override its recommendations.

Interpretability also enhances patient safety by enabling clinicians to detect potential failure modes in AI policies. Suppose a policy begins recommending inappropriate treatment, such as excessive fluid administration in a patient with pulmonary edema. With CQL, clinicians can use LEG analysis to identify which features are driving the erroneous recommendation and potentially adjust the input state by correcting a mislabeled blood pressure reading or overriding the recommendation. With DQN, the lack of interpretable feature importance makes it nearly impossible to diagnose why the policy is failing, leaving clinicians with a binary choice: blindly trust the system or abandon it entirely. This diagnostic capability is essential for safe deployment of AI in high-stakes medical environments.

Based on our findings, we propose the following guidelines for selecting RL algorithms for clinical decision support. Interpretability should be prioritized alongside performance. When multiple algorithms achieve similar outcomes, the algorithm with the strongest, most clinically coherent interpretability metrics, such as LEG saliency scores, should be selected. Conservative offline RL methods like CQL should be used as the default choice because conservatism enhances both safety by avoiding out-of-distribution actions and interpretability by biasing toward simpler, threshold-based decision rules. Unconstrained online RL methods like DQN should be avoided for clinical deployment because while such methods may perform well in simulation, their lack of interpretability renders them unsuitable for regulatory approval and clinical trust. Interpretability should be validated quantitatively using gradient-based or perturbation-based methods such as LEG and SHAP to measure feature importance and ensure that top-ranked features align with medical knowledge.

4.4 Limitations and Caveats

While our study provides valuable insights into the performance-interpretability trade-off in offline RL for sepsis treatment, several limitations warrant discussion.

A critical limitation of our study design is that DQN's training paradigm creates ambiguity in our "offline vs. online" comparison. As detailed in Section 2, DQN was trained online by interacting with the Gym-Sepsis simulator for 100,000 timesteps, yet lacks both offline-specific safeguards, such as CQL's conservatism penalty, and online-specific architectural innovations, such as DDQN-Attention's multi-head attention, DDQN-Residual's skip connections, and SAC's entropy regularization. This creates a confound. We cannot cleanly attribute DQN's poor interpretability with a maximum saliency of 0.069, which is 600-fold weaker than CQL, and inferior high-SOFA

performance at 84.3% survival vs. 88.5% for CQL and 90.5% for DDQN-Attention to either the absence of offline conservatism, the absence of architectural sophistication, or an interaction between training paradigm and network design.

From our results, we can conclude that CQL achieves superior interpretability and competitive performance when trained offline with conservatism. DDQN-Attention/Residual and SAC achieve marginally better survival when trained online with architectural enhancements, but with 11-fold weaker interpretability than CQL. Vanilla DQN performs poorly on both dimensions when trained online without enhancements. However, we cannot conclusively determine whether DQN's failures stem from its online training exploring suboptimal regions of the state-action space or its vanilla architecture lacking regularization mechanisms. A proper ablation study would require training DQN offline on the same 2,105-episode dataset with approximately 20K transitions used by BC and CQL, isolating the effect of training paradigm from architecture.

Despite this ambiguity, our practical recommendation remains clear. For clinical deployment, both offline training to avoid patient risk during learning and conservative constraints to ensure interpretable, safe policies are essential. DQN's dual deficiencies of online training requirement and poor interpretability render it unsuitable regardless of which factor is causally dominant. Future work should compare truly offline-trained DQN with CQL to isolate the value of conservatism within the offline paradigm, enabling sharper mechanistic conclusions about what drives interpretability.

A methodological limitation of our evaluation is that each algorithm was evaluated on an independently sampled set of 500 episodes from the MIMIC-III-derived starting state distribution, rather than on identical patient trajectories. This resulted in different SOFA-stratified sample sizes across algorithms, with 185 high-SOFA patients for DQN, 191 for CQL, and 211 for BC, reflecting the stochastic episode initialization process. While all algorithms were evaluated using the same simulator and sampling distribution, the specific patient cases differed between algorithms, precluding direct patient-level comparisons. However, with 500 episodes per algorithm, a sample size substantially larger than typical clinical trials, performance metrics should converge to distribution-level expectations. The observed performance differences with survival rates spanning 94.0% to 95.4%, a 1.4 percentage point range, are consistent with both genuine algorithmic differences and expected sampling variability from independent 500-episode draws. Formal statistical testing, such as permutation tests or bootstrapped confidence intervals, could quantify uncertainty but was not performed in this study. Future work should implement fixed-seed evaluation protocols to enable exact pairwise comparisons and stronger claims about relative performance. Nonetheless, our primary conclusions that CQL achieves substantially stronger interpretability at 40.06 vs. 0.069–3.57 max saliency while maintaining competitive survival rates remain robust to this limitation, as the interpretability differences span 3 orders of magnitude and are unlikely to be explained by episode sampling variance.

Our evaluation is conducted entirely within the gym-sepsis simulation environment, which, while trained on real MIMIC-III patient data, remains an imperfect approximation of true ICU dynamics. This sim-to-real gap introduces several sources of uncertainty. The simulator's transition dynamics and outcome model are learned from observational data, which may not accurately capture causal relationships between treatments and outcomes. For example, the simulator might overestimate treatment benefits if sicker patients received more aggressive care in the training data. A striking illustration of this limitation is that the random policy achieves 95.0% survival, matching

or exceeding several trained policies, including the guideline-based heuristic at 94.6%, BC at 94.2%, and CQL at 94.0%. This counterintuitive result suggests the simulator's outcome model is largely insensitive to treatment choices, likely because the underlying MIMIC-III data reflects a population where most patients survived regardless of the specific treatments they received. The simulator may have learned to predict survival based primarily on patient characteristics rather than treatment actions, effectively decoupling outcomes from interventions. This does not imply that treatment is clinically irrelevant; real sepsis outcomes depend critically on timely, appropriate interventions. Rather, it highlights that our simulator cannot reliably distinguish good from bad treatment policies based on survival alone, potentially masking performance differences that would emerge in practice. Offline policy evaluation within a simulator compounds these uncertainties because our survival estimates reflect how policies perform in a model of reality, not reality itself. While offline policy evaluation is standard practice in offline RL research,[15] bridging this gap requires prospective evaluation methods such as off-policy evaluation on real-world EHR data, semi-synthetic benchmarks that combine real data with learned dynamics, or ultimately, prospective clinical trials. Our results should therefore be interpreted as a proof-of-concept demonstrating relative interpretability differences across algorithms rather than definitive evidence of clinical superiority.

Our sparse terminal reward of +15 for survival, −15 for death, and 0 for intermediate steps captures the primary clinical objective of patient survival but oversimplifies the multifaceted goals of sepsis management. This reward design does not account for intermediate treatment costs such as medication side effects and ICU resource utilization, long-term quality of life such as cognitive impairment or organ damage post-discharge, or clinician workload. Consequently, our learned policies may recommend treatment sequences that maximize short-term survival at the expense of these unmodeled factors. For example, a policy might aggressively administer vasopressors to maintain blood pressure, potentially increasing survival but causing downstream cardiac complications. Future work should explore shaped reward functions that incorporate domain knowledge about acceptable treatment trade-offs, though designing such rewards without introducing unintended biases remains a significant challenge. The interpretability advantages of CQL observed under our sparse reward may or may not generalize to more complex reward structures.

Our LEG interpretability analysis relies on local linear approximations of the Q-function, which may not fully capture non-linear interactions among features. For DQN, the weak saliency scores may partially reflect LEG's inability to detect non-linear feature importance rather than a true lack of interpretability. Alternative interpretability methods, such as SHAP values[32], which account for feature interactions via Shapley values or attention-based mechanisms that explicitly model feature weighting, might reveal additional structure in DQN's decision-making. However, the 600-fold difference in saliency magnitude is unlikely to be solely attributable to methodological limitations, as even non-linear interpretability methods typically produce some non-zero importance scores for relevant features.

Our study focuses on 3 specific offline RL algorithms, including BC, CQL, and DQN, and does not explore other promising methods, such as Implicit Q-Learning[33], Decision Transformer[34], or model-based offline RL. These methods may offer different performance-interpretability trade-offs, and future work should extend our LEG analysis framework to a broader set of algorithms. Additionally, our choice of CQL hyperparameters with $\alpha = 1.0$ follows default recommendations in the d3rlpy library but may not be optimal for interpretability. Systematic tuning of α to maximize both

performance and interpretability could further improve CQL's clinical suitability.

Our evaluation uses a relatively small sample of 10 states per algorithm for LEG analysis, selected uniformly across SOFA severity levels. While these states were chosen to be representative, a more comprehensive analysis covering hundreds of states across diverse patient subpopulations stratified by age, comorbidities, and infection source would strengthen confidence in the generalizability of our interpretability findings. Additionally, our interpretability assessment focuses on feature importance as measured by saliency but does not address other dimensions of interpretability, such as action consistency, meaning whether the policy makes similar decisions in similar states, or counterfactual reasoning, meaning what would happen if a specific feature were different. Future work should incorporate these additional interpretability criteria for a more holistic evaluation.

Our study does not address the important question of how interpretability affects clinical outcomes when AI recommendations are actually deployed. It is possible that highly interpretable policies like CQL improve clinician trust and adoption, leading to better adherence and ultimately better patient outcomes. Alternatively, interpretability might have little impact on outcomes if clinicians override AI recommendations regardless of transparency. Prospective human-in-the-loop studies, where clinicians interact with CQL and DQN policies in simulated or real clinical scenarios, are needed to assess the causal effect of interpretability on decision-making and patient safety.

4.5 Future Directions

Our work opens several promising avenues for future research. Extending our evaluation to real-world clinical data is critical. Prospective evaluation using real-world EHR data from multi-center ICU cohorts such as the eICU Collaborative Research Database[35] would validate whether offline RL's interpretability advantages and comparable performance to online methods persist in diverse clinical settings with varying patient populations and treatment protocols.

Investigating the causal relationship between interpretability and clinical outcomes through randomized human-in-the-loop experiments is essential. Such studies would randomize clinicians to receive recommendations from high-interpretability algorithms like CQL or low-interpretability algorithms like DQN and measure differences in recommendation adherence, decision-making time, and patient outcomes. If interpretability causally improves clinician trust and decision quality, this would provide strong evidence for prioritizing interpretable offline RL algorithms in clinical AI development.

Extending outcome metrics beyond acute survival to include long-term neurological sequelae represents a critical frontier for sepsis RL research. Beyond its immediate life-threatening effects, sepsis has profound implications for long-term neurological health through mechanisms including blood-brain barrier disruption[7], microglial activation and oxidative stress[8], and accelerated neurodegenerative processes in vulnerable populations.[9] Patients with pre-existing neurological conditions face particularly grave consequences, as sepsis episodes can trigger irreversible cognitive decline and worsen underlying pathologies such as Alzheimer's disease. This bidirectional relationship between sepsis and neurological disorders highlights the critical need for interpretable, effective treatment algorithms like CQL that can optimize acute sepsis management while potentially mitigating long-term neurological sequelae. Future RL frameworks should incorporate composite reward functions that balance acute hemodynamic stabilization with neuroprotective

treatment strategies, potentially integrating biomarkers of BBB integrity, inflammatory burden, and cognitive trajectories. Such multi-objective optimization would require prospective cohort studies linking acute treatment decisions to long-term neurocognitive outcomes, an endeavor where interpretable algorithms like CQL may prove essential, enabling clinicians to understand not only which treatments maximize short-term survival but also why certain interventions may mitigate long-term neurological damage. This research direction aligns with growing recognition that critical illness survivorship extends far beyond hospital discharge, encompassing quality of life, functional independence, and cognitive preservation as equally vital endpoints.

5. Conclusion

This study demonstrates that offline RL algorithm selection profoundly impacts policy interpretability independent of performance in the gym-sepsis simulator. Our systematic LEG-based benchmark shows that Conservative Q-Learning achieves substantially stronger saliency magnitude than vanilla DQN, spanning multiple orders of magnitude (40.06 vs. 0.069, representing a 600-fold difference), while achieving comparable survival outcomes. While all algorithms achieve similar overall survival rates ranging from 94.0% to 95.4%, CQL substantially outperforms DQN on high-severity patients with SOFA ≥ 11 (88.5% vs. 84.3%), representing a clinically meaningful performance gap. CQL's strong, clinically coherent saliency patterns emphasizing blood pressure and lactate align with Surviving Sepsis Campaign guidelines[4] and provide quantitative feature importance suitable for regulatory review.[12]

Our findings within this single simulator suggest that conservative offline RL methods warrant consideration for safety-critical healthcare applications where interpretability is essential. CQL's conservative penalty term biases the Q-function toward values supported by the training dataset, inheriting the behavioral policy's interpretable structure based on threshold-based heuristics.[17] This interpretability-by-design mechanism offers an alternative to post-hoc explainability methods. The clinical urgency extends beyond acute survival to preventing long-term neurological sequelae, as sepsis triggers blood-brain barrier disruption, neuroinflammation, and accelerated neurodegeneration.[7,8,30] Interpretable policies like CQL enable clinicians to validate whether treatment recommendations account for these complex pathophysiological interactions.

Several limitations should be acknowledged. Our conclusions are based on a single sepsis treatment simulator with known constraints, including high baseline survival (approximately 94–95% across all policies, including random) and learned dynamics that may not capture true causal treatment effects. The evaluation used independently sampled episodes rather than identical patient trajectories, precluding direct patient-level comparisons. Additionally, LEG analysis relies on local linear approximations that may not fully capture non-linear feature interactions. Claims about broader applicability to other clinical domains or real-world deployment require prospective validation through multi-center clinical trials, human-in-the-loop studies with clinicians, and evaluation on real EHR data.[35]

Future work should pursue prospective clinical validation, integrate domain knowledge into CQL training through feature-aware regularization, develop healthcare-specific interpretability metrics, including stability, parsimony, and actionability, and conduct human-in-the-loop experiments to establish causality between interpretability and clinical decision quality.[29] By prioritizing interpretability alongside performance in algorithm design, we can develop clinical AI

systems that earn the trust and adoption of clinicians while improving patient care and potentially mitigating long-term neurological sequelae in sepsis survivors.

AI Use Declaration

During the preparation and revision of this manuscript, the authors used the following artificial intelligence–assisted tool(s): [name of AI tool(s), e.g., ChatGPT (OpenAI), Grammarly, or other language-support tools], solely for the purpose of improving the clarity, readability, and linguistic quality of the text and/or assisting in the preparation of figures or visual materials, where applicable.

All scientific content, including the study design, data acquisition, data analysis, interpretation of results, discussion, and conclusions, was conceived, generated, and critically verified by the authors. The authors take full responsibility for the accuracy, originality, and integrity of the published work.

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References

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